

Wave 5.2R Stage 13 Synthetic And Weak-Form Oracle Lane Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Outcome

Stage 13 completed all 10 / 10 analytical-oracle entries with zero execution failures. All ten predeclared synthetic gates passed, deterministic replay was exact, test-label dependence was absent, and no real-data model was promoted.

The main positive result is not a new predictor. It is a certified weak-form harmonic diagnostic that remains specific to the declared oscillator law while avoiding the severe noise amplification observed when a second derivative is estimated pointwise.

Campaign Integrity

- Campaign: wave52r_stage13_synthetic_weak_form_oracle_lane_2026_07_29
- Output:
output/training_campaigns/2026-07-29-23-33-57_wave52r_stage13_synthetic_weak_form_oracle_lane_2026_07_29
- Completed entries: 10
- Execution failures: 0
- Certified synthetic cases: 10
- Rejected cases: 0
- Oracle conditions: 64 training-domain H04 conditions
- Test curves checked for provenance only: 97
- Test-label dependence: false
- Real-data promotion allowed: false

Certification Matrix

ID	Experiment	Primary result	Gate	Decision
C00	exact H04 reconstruction	1.53e-16 max error	$\leq 1e-10$	certified
H01	harmonic injection	1.36e-15 normalized error	≤ 0.02	certified
H02	harmonic omission	3.63e12 error ratio	≥ 5	certified
C01	coefficient-surface perturbation	1.83e-14 normalized error	≤ 0.02	certified
M01	misspecified anchor	2.55e-14 normalized error	≤ 0.02	certified
Q01	compliance nonlinearity	5.84e-16 normalized error	≤ 0.05	certified
P01	pointwise oscillator residual	103.27 maximum residual	finite baseline	certified baseline

ID	Experiment	Primary result	Gate	Decision
W01	weak oscillator residual	0.000640	<= 0.02	certified
D01	sampling-density stress	128 samples	<= 256	certified
N01	wrong-law and shuffled controls	141.08 rejection ratio	>= 10	certified

The near-machine-precision recovery results are expected because these cases certify internally consistent synthesis and projection paths. They validate implementation correctness, not real reducer mechanics.

Harmonic Injection And Omission

The frozen H04 core basis contains orders 1, 3, 39, 40, 78, 81, 156, 162, and 240. Stage 13 injected the omitted order 2 with known sine and cosine coefficients.

The extended basis recovered both injected coefficients with normalized error 1.36e-15. Removing order 2 increased mean curve RMSE from 2.42e-17 deg to 0.003630 deg, a ratio of 3.63e12. This proves that the implemented diagnostic can identify an omitted resolvable harmonic under the ideal synthetic contract.

Weak-Form Versus Pointwise Residual

The matched signal satisfies $u''(\theta) + 5^2 u(\theta) = 0$.

The pointwise lane estimates u'' from corrupted samples. The weak lane moves the derivative onto smooth periodic test functions and integrates against u . The comparison uses identical signals, seeds, densities, and noise.

Samples	Noise	Pointwise residual	Weak residual
2048	0	1.96e-05	5.56e-06
2048	0.01	103.273	0.000209
1024	0.01	26.591	0.000153
512	0.01	6.421	0.000377
256	0.01	1.698	0.000640
128	0.01	0.320	0.001486

Weak-form residuals were lower in all 12 / 12 non-zero-noise comparisons at densities of at least 256. At the predeclared boundary of 256 samples and noise 0.01, the weak residual is 0.000640, more than thirty times below the 0.02 gate.

The pointwise residual increases with density under fixed per-sample noise because the second-difference operator amplifies high-frequency corruption by the inverse square of angular spacing. The weak residual does not differentiate the measured signal and therefore remains stable.

Specificity And Observability

At 256 samples and noise 0.001

- correct-law weak residual: 0.0003598 ;
- wrong-order residual: 0.0868583 , or 241.41 times larger;
- shuffled-angle residual: 0.0507615 , or 141.08 times larger.

The minimum negative-control rejection ratio exceeds the required 10 by more than one order of magnitude. The weak diagnostic is therefore not merely small for every input. All declared densities down to 128 satisfy the weak residual threshold at noise 0.01 . Stage 13 consequently records 128 as the minimum certified density inside the tested range. It does not extrapolate below that range.

What The Result Establishes

Stage 13 establishes that

- Fourier synthesis and coefficient projection are internally consistent;
- injected and omitted harmonics are detectable under an observable basis;
- known coefficient, anchor, and compliance perturbations are recoverable;
- the weak harmonic residual is substantially more noise-robust than the pointwise second-derivative residual;
- wrong-law and shuffled-angle controls are rejected;
- the full result is exactly reproducible with the frozen seed and contract.

What The Result Does Not Establish

Stage 13 does not establish that

- order 2 or the synthetic compliance law is a true reducer mechanism;
- the weak residual will improve real measured-curve prediction;
- a physical coefficient is identifiable from setpoints alone;
- H04, K01, or any PINN should be promoted;
- synthetic success satisfies Stage 14 tournament entry.

Any weak-form training loss must enter a later isolated real-data matched control with leakage, causality, curve-first, and deployment gates.

Decision And Next Step

The Stage 13 lane is certified_for_synthetic_use . No real-data winner is declared and the accepted model registries remain unchanged.

Stage 14 may now begin the Cross-Formulation Forward Tournament. Only candidates that already passed their isolated real-data gates may enter. Stage 13 evidence can support implementation confidence and observability requirements, but it cannot make an otherwise unqualified candidate eligible.

Physics-integrated Wave 6 remains closed.

For reproducibility, the campaign uses split signature

c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16 , seed 314159 , H04 core orders and curve scale, immutable campaign and validation directories, explicit noise and density grids, and exact deterministic replay.